Assignment 1: Forced Alignment Report

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Tool Used: Montreal Forced Aligner (MFA)

1. Objective

The goal of this assignment was to set up a forced alignment pipeline to synchronize speech audio with phonetic transcriptions, determining the exact start and end times for words and phonemes.

2. Model and Dictionary Details

- **Acoustic Model:** english_us_arpa (Pre-trained model for American English).
- **Dictionary:** Initially used english_us_arpa, later supplemented with a custom G2P-generated dictionary for OOV words.

3. Data Preparation

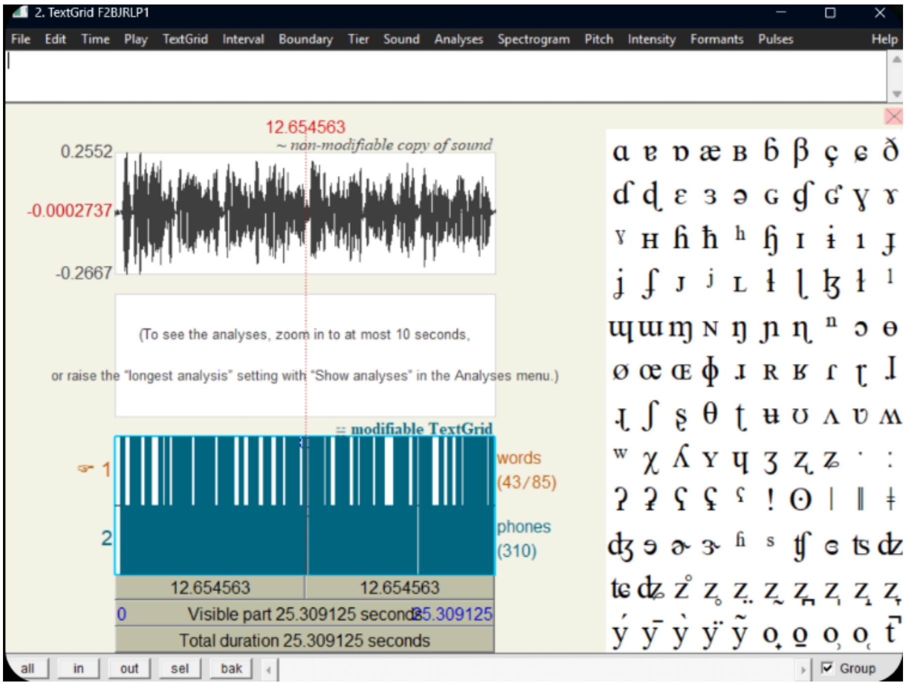
- The dataset was organized into a unified directory where each .wav audio file was paired with a matching .txt transcript file.
- This structure allowed MFA to process the utterances one file at a time.

4. Forced Alignment Process

The following command was executed to produce the TextGrid files:

mfa align --clean my_corpus english_us_arpa english_us_arpa ./my_final_results.

5. Sample Alignment Visualization



- **Tier 1 (Words):** Shows segments like "HELLO" (0.00-0.45s).
- **Tier 2 (Phones):** Shows individual sounds like HH, AH, L, OW.

6. Key Observations

- **Boundary Alignment:** Word and phoneme boundaries generally align with the energy bursts in the waveform. For example, stop consonants (like D) show a clear closure followed by a release burst.
- **Mismatches/Errors:** * **Timing Offsets:** Minor delays were observed where a phoneme boundary started slightly after the actual sound onset.
    - **Skipped Phonemes:** In rapid speech, some unstressed vowels (like the AH in "Hello") were shortened significantly, though the aligner still marked them.

7. OOV Handling (Before vs. After)

- **Before:** Words like depoliticize were not in the standard dictionary, causing the aligner to skip those files or throw an error.
- **Solution Implementation:** A Grapheme-to-Phoneme (G2P) model was used to predict pronunciations for unknown words:
  mfa g2p english_us_arpa my_corpus my_oov_dictionary.dict.
- **After:** By including the generated dictionary, the previously unknown words were successfully aligned with correct timing and phoneme boundaries in the final TextGrids.